Fluctuation and Noise Letters

A Controlled Benchmark for Statistical Detection of Noise-Induced Transitions in Nonlinear Stochastic Dynamics

Nikita Krutikov
HSE University
Moscow, Russia

Petr Lukianchenko*
HSE University
Moscow, Russia
plukyanchenko@hse.ru

Noise-induced transitions are central to stochastic dynamics, but statistical detection benchmarks often use externally inserted shifts. We study a controlled alternative: trajectories of a nonlinear stochastic system where events are labeled by transitions between state-space regions after simulation. The task is therefore transition-time detection rather than ordinary parameter-break detection. Classical tests, window-based procedures, and learned detectors are evaluated under one protocol with independent training, calibration, and test trajectories. In the frozen four-scenario benchmark, classical homogeneity tests lead the sparse low-D white-noise case. Under the primary 25-step tolerance, learned window-based detectors lead the evaluated $D = 1.0$ cases and the strict-margin $D = 1.5$ white-noise case, whose ranking is sensitive to wider tolerances. At $D = 1.0$ with white noise, Transformer has the highest aggregate mean F1, but its paired difference from LSTM is not statistically significant. A same-noise Transformer diagnostic shows that noise-specific training alone does not produce a universal colored-noise neural advantage under the evaluated pipeline.

Keywords: Change point detection; stochastic dynamics; noise-induced transitions; colored noise; machine learning

1. Introduction

Noise-induced transitions are a natural object of study in stochastic dynamics: random fluctuations can move a trajectory between regions of state space even when the underlying dynamical rule is fixed. From a statistical point of view, the question is how accurately such transition times can be detected from an observed series. Standard change point detection (CPD) simulations usually test externally inserted shifts in mean, variance, or distribution; they do not cover events generated by the stochastic process itself.

This paper studies that complementary setting. We generate trajectories from a nonlinear stochastic dynamical system and label events by a fixed crossing rule

*Corresponding author.

between state-space regions. The generating mechanism is not changed at the event time; the label marks a transition of the trajectory between regions. Thus the benchmark is designed for transition-time detection in stochastic dynamics, not for structural-break detection in the usual parameter-change sense. Rather than introducing a new detector or escape-rate theory, we use this dynamics-generated process to compare existing detectors under fixed training, calibration, testing, and matching rules.

The study makes four contributions. It defines a reproducible transition-time benchmark with a specified generator, level-crossing labels, metadata, and independent train/calibration/test splits. It evaluates classical tests, segmentation and window-based procedures, and learned detectors under the same matching protocol. It reports D-dependent rankings, including the tolerance-sensitive $D = 1.5$ case, and uses a same-noise Transformer diagnostic to delimit colored-noise neural-advantage claims.

2. Related Work

Change point detection (CPD) methods are usually grouped into online and offline formulations; the offline setting used here estimates event locations after the full series has been observed [1]. Surveys emphasize that benchmark conclusions depend on detector assumptions, calibration, post-processing, data generation, annotation, and matching rules [2–4]. Most simulations prescribe the change structure directly, for example by placing shifts in a piecewise signal or concatenating distinct segments [5–8]. Here the labels come from a simulated stochastic trajectory: their number and spacing are consequences of the dynamics and noise realization.

This distinction separates the benchmark from SDE-based detection models. Recent work has used latent stochastic differential equations as a model class for change point detection [9]. Here the stochastic dynamics are not the detector. They define the transition process on which existing classical, window-based, and learned detectors are evaluated under one fixed protocol.

The stochastic-dynamics motivation is rooted in noise-induced motion between regions of state space. Classical work on random perturbations, escape problems, and sample paths describes how fluctuations can drive transitions in nonlinear systems [10–12]. Within *Fluctuation and Noise Letters*, noise-induced escape from the Lorenz attractor has been studied as an escape problem in nonlinear dynamics [13]. Colored noise is also a natural part of this setting rather than only a robustness stress test: FNL studies have examined the role of colored noise in stochastic-system behavior [14], and recent work has used deep learning to characterize colored-noise time-series patterns [15]. These papers motivate treating the colored-noise diagnostic in this article as a boundary on neural claims, not as a generic machine-learning add-on.

The compared methods cover standard CPD families: CUSUM-type, rank-based, homogeneity, and structural-break tests [16–20]; segmentation and local-window methods [5,21,22]; and learned detectors trained from labeled windows [23]. Because learned detectors use generated labels, the benchmark keeps training, calibration, and test trajectories independent.

3. Data-Generating Process

The benchmark generator is the discrete-time stochastic map

$$x_{t+1} = x_t + \sin(x_t)\Delta t + \sqrt{D}\varepsilon_t, \tag{1}$$

where the initial state is close to zero and $\sin(x_t)$ gives a nonlinear periodic drift. The innovation sequence ε_t is generated for a fixed length, centered, and scaled to unit empirical variance when possible. The map is motivated by stochastic dynamics, but it is not treated as a strict Euler–Maruyama discretization of a continuous-time diffusion: the implementation applies the $\sqrt{D}$ factor shown above and does not add a separate $\sqrt{\Delta t}$ factor. Thus D controls the perturbation scale of this discrete benchmark source.

Colored innovations are generated by Fourier-domain spectral shaping of a Gaussian white sequence [24, 25]. For a length-N draw, the real Fourier transform Z_k is multiplied by

$$a_c(f) = \begin{cases} 1, & \text{white}, \\ f^{-1/2}, & \text{pink}, \\ f^{-1}, & \text{Brownian/red}, \\ f, & \text{violet}, \\ f^{1/2}, & \text{blue}, \end{cases} \qquad f > 0. \tag{2}$$

The zero-frequency component is not used for singular colored families. The weights are normalized by their root mean square before the inverse transform; the real sequence is then centered and scaled. Thus the innovation family controls temporal correlation, not empirical variance scale.

3.1. *Transition Labels*

Change points in this benchmark are event labels, not parameter breaks in the generator. Let L_t denote the integer level assigned to x_t/π by the canonical level-mapping rule. The binary label at time t is one when $L_t \neq L_{t-1}$ and zero otherwise. The same stochastic map therefore generates the whole path, while the level-crossing rule converts that path into transition-time labels. A rule that directly recomputes L_t from x_t would reproduce the annotation mechanism; it is not treated as a competing CPD detector. The compared methods are evaluated as general-purpose detectors applied to the trajectory without using the precomputed level sequence.

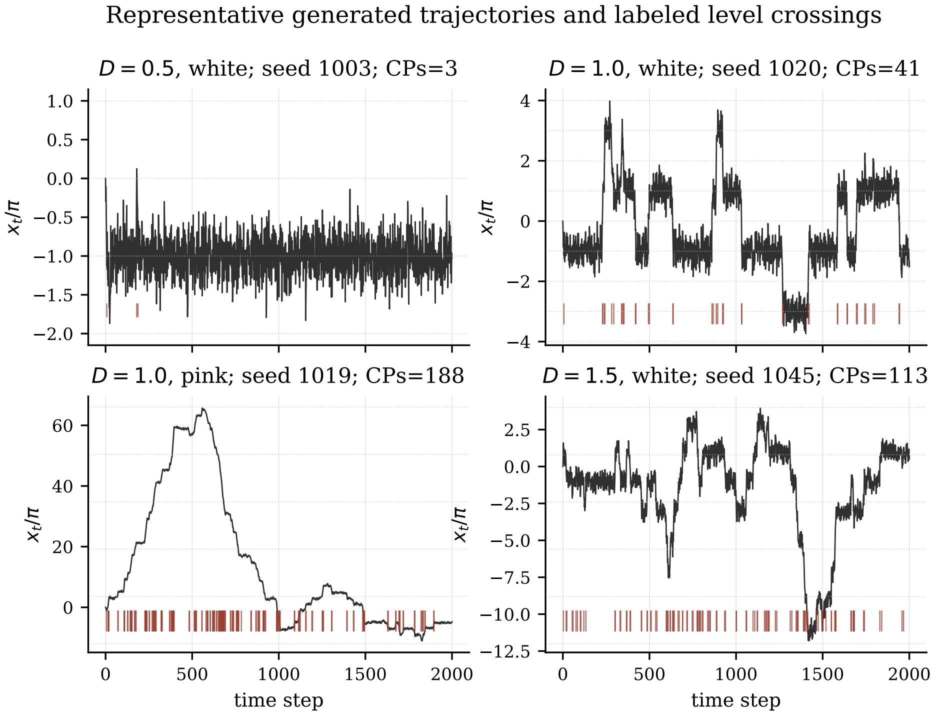

Fig. 1. Representative trajectories and labeled level crossings under the four frozen benchmark scenarios. Red ticks indicate true labels; larger D or correlated innovations create denser transition neighborhoods.

3.2. *Transition Properties*

The parameter D controls how often the trajectory leaves one level region. In a diagnostic grid with $\Delta t = 1.0$, white innovations, length 10000, and 30 series per setting, $D = 0.5$ produced 6.4 change points per series on average, with mean positive holding time about 936 steps. For $D = 1.0$ and $D = 1.5$, the corresponding values were about 197/50 and 576/17. Thus the $D = 1.5$ case is dense relative to the 25-step matching tolerance and must be read with the robustness check.

Because colored innovations change temporal correlation structure, colored-noise comparisons are treated as separate experimental settings rather than direct quantitative extensions of the white-noise cases.

The generator outputs $(x_{0:T}, y_{0:T})$, both of length $T + 1$, with zero-based indexing. No-change cases use an all-zero label vector or an empty change-point list. Reportable runs store the dataset source, generation parameters ($T, \Delta t, D$, noise type, seed), and source path.

Table 1. Final learned-detector configurations used in the benchmark.

Method	Window	Architecture	Test post-processing
CatBoost	50	14 statistical window features; 500 trees, depth 6, learning rate 0.05	threshold 0.57; NMS 50
LSTM	100	two LSTM layers, hidden size 256, dropout 0.2; linear logit head	threshold 0.15; NMS 10
GRU	100	two GRU layers, hidden size 128, dropout 0.2; linear logit head	threshold 0.15; NMS 10
Transformer	100	$d_{\mathrm{model}} = 256$, 8 heads, 6 encoder layers, feed-forward size 512, dropout 0.1; mean pooling and linear logit head	threshold 0.15; NMS 10

4. Methods

All detectors use a common interface: a one-dimensional series and seed are mapped to zero-based predicted change points, optional per-timestep scores, and method metadata. The shared evaluation layer then applies the same matching margin and metrics to every method.

The classical group contains distributional and structural-break tests adapted to the generated trajectories. Pettitt's rank-based test, SNHT, Buishand-type homogeneity tests, and the Chow test provide interpretable baselines for level or regression-structure changes. CUSUM is included as a sequential mean-change detector derived from Page's cumulative-sum idea [16]. Additional segmentation-style baselines in the executable registry include MDL, SWAB, Gaussian-process-based detection, and a Nyblom test variant. These methods do not learn from the training split, but their outputs are converted to the same change-point-list representation.

The learned group contains CatBoost, recurrent neural detectors, and a Transformer detector. All learned detectors operate on normalized sliding windows and return a local probability or logit for the window centre. Dense scores are then thresholded and converted to change-point lists by non-maximum suppression (NMS). The final benchmark configuration is summarized in Table 1; the neural detectors are trained on all five innovation families with D sampled from $[0.2, 1.0]$ and $\Delta t \in \{0.5, 1.0, 2.0\}$. The $D = 1.5$ test scenario is therefore an extrapolation test for the learned detectors with respect to the training D-range.

The comparison separates method-specific scoring from shared evaluation. Only final predicted change-point lists and, when available, dense score arrays are passed to metric computation.

5. Experimental Protocol

The experiments follow a fixed data-generation contract. The current manuscript tables use `algoritms/data_utils.py`, the executable mirror of the notebook source. Run metadata record length, time step, D, noise type, seed, dataset source, and source path. Allowed innovation families are white, pink, Brownian, violet, and blue, each centered and scaled before entering the map.

The benchmark uses independent generated trajectories, not timestep-level splits: training seeds 0–99, calibration seeds 5000–5019, and test seeds 1000–1049. The four frozen test scenarios are $D = 0.5$ white, $D = 1.0$ white, $D = 1.0$ pink, and $D = 1.5$ white, all with length 2000 and $\Delta t = 1.0$.

Methods that require fitting use only the training split; thresholds and other selection decisions are tuned on calibration data and frozen before test evaluation. Classical methods are evaluated under the same matching and metric definitions as learned detectors.

Change points are indexed from zero. The primary benchmark counts a predicted change point as a match when it lies within 25 timesteps of an unmatched true change point. This margin is an evaluation tolerance rather than a detector hyperparameter, and is distinct from the non-maximum-suppression distance used inside some learned detectors. The matching is one-to-one within a series. The reported metrics are precision, recall, F1 score, and mean absolute localization error for matched detections. For methods that return dense scores, ROC-AUC and PR-AUC are also reported. When a method detects no change points, its prediction is represented by an empty list rather than a missing value. At the series level, a trajectory with no true and no predicted change points is assigned precision, recall, and F1 equal to 1; a trajectory with no true change points but at least one prediction is counted as a false-positive case and receives zero precision, recall, and F1.

5.1. *Choice of Matching Margin and Robustness*

The primary margin of 25 timesteps matches the frozen benchmark outputs used for the main leaderboard. To check whether the qualitative conclusions depend on this tolerance, all methods were rerun with frozen weights and unchanged post-processing, and the same predictions were evaluated at matching margins 25, 50, and 100. This check is especially important for the $D = 1.5$ scenario, where the diagnostic mean holding time is shorter than the primary matching tolerance. No model was retrained in this robustness pass. Table 2 reports the leading method in each scenario under each margin.

The low-D result is stable: SNHT remains the leading method for $D = 0.5$ with white noise at all three margins. The two $D = 1.0$ scenarios also preserve the same leading method, although wider margins increase the F1 of CUSUM. The $D = 1.5$, white-noise scenario is more sensitive to the chosen tolerance: the Transformer detector leads under the primary, stricter margin, whereas CUSUM leads when detections within 50 or 100 timesteps are counted as matches. This indicates that the large-D ranking partly reflects localization strictness, not only event-detection ability.

Table 2. Leading mean F1 under alternative matching margins. The detector outputs and thresholds are fixed; only the evaluation tolerance changes.

Scenario	Margin 25	Margin 50	Margin 100
$D = 0.5$, white	SNHT 0.725	SNHT 0.725	SNHT 0.725
$D = 1.0$, white	Transformer 0.698	Transformer 0.723	Transformer 0.790
$D = 1.0$, pink	Transformer 0.566	Transformer 0.589	Transformer 0.610
$D = 1.5$, white	Transformer 0.643	CUSUM 0.666	CUSUM 0.692

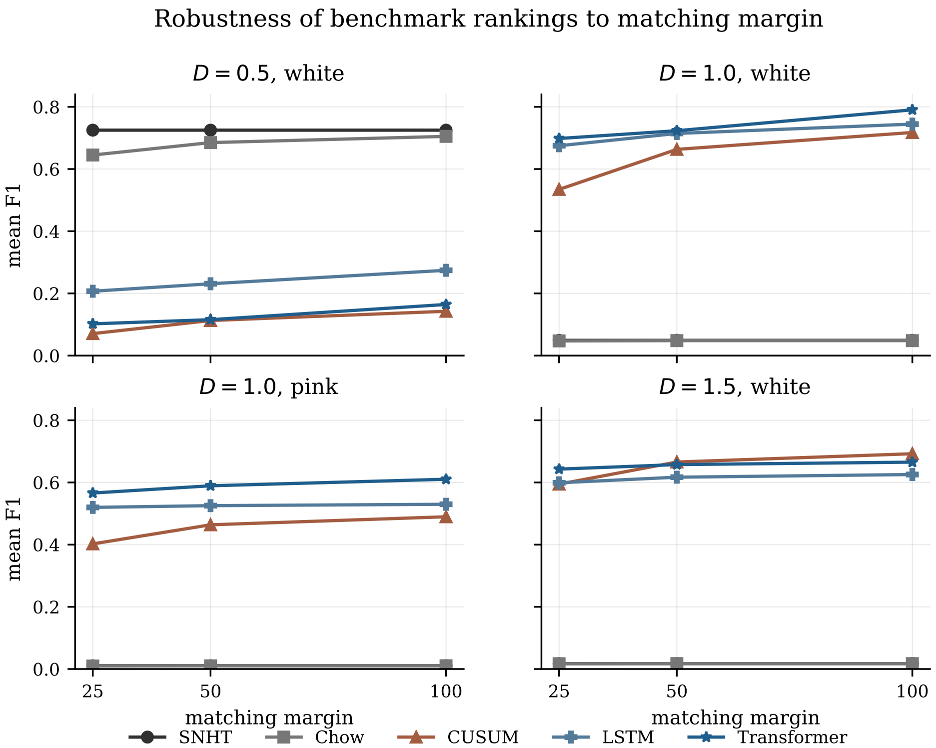

Fig. 2. Sensitivity of selected benchmark rankings to the matching margin. Detector outputs and post-processing are fixed; only the evaluation tolerance changes. The low-D and $D = 1.0$ conclusions are stable, while the $D = 1.5$ white-noise scenario shows the CUSUM–Transformer crossover under wider matching tolerances.

6. Results

This section reports the frozen four-scenario benchmark and the same-noise Transformer diagnostic. Validation checks confirmed the expected test seeds 1000–1049 and reproduced the stored scenario summaries from per-series outputs under the

Table 3. Top five methods by mean F1 in each frozen benchmark scenario. Precision (P), recall (R), F1, mean absolute localization error (MAE), ROC-AUC (ROC), and PR-AUC (PR) are reported when available.

Scenario	Method	P	R	F1	MAE	ROC	PR
D=0.5, white	SNHT	1.000	0.641	0.725	1.360		
D=0.5, white	Chow	0.920	0.561	0.645	4.520	0.684	0.533
D=0.5, white	GRU	0.322	0.230	0.253	3.180	0.918	0.551
D=0.5, white	CatBoost	0.355	0.213	0.243	4.280	0.650	0.211
D=0.5, white	LSTM	0.237	0.209	0.207	2.320	0.866	0.424
D=1, white	Transformer	0.774	0.641	0.698	3.560	0.815	0.799
D=1, white	LSTM	0.840	0.569	0.675	3.350	0.813	0.811
D=1, white	GRU	0.913	0.511	0.650	3.690	0.847	0.850
D=1, white	CUSUM	0.462	0.645	0.534	9.790	0.483	0.467
D=1, white	CatBoost	0.921	0.325	0.479	2.290	0.733	0.788
D=1, pink	Transformer	0.916	0.410	0.566	3.330	0.896	0.972
D=1, pink	GRU	0.987	0.384	0.551	2.610	0.922	0.981
D=1, pink	LSTM	0.981	0.354	0.520	2.100	0.870	0.966
D=1, pink	CUSUM	0.794	0.270	0.402	7.500	0.474	0.783
D=1, pink	CatBoost	0.850	0.124	0.216	1.610	0.666	0.915
D=1.5, white	Transformer	0.964	0.483	0.643	3.210	0.810	0.960
D=1.5, white	LSTM	0.956	0.438	0.599	3.590	0.807	0.960
D=1.5, white	CUSUM	0.845	0.461	0.595	7.980	0.458	0.837
D=1.5, white	GRU	0.969	0.278	0.429	3.280	0.806	0.959
D=1.5, white	CatBoost	0.991	0.206	0.341	2.100	0.828	0.968

primary 25-step matching margin. Table 3 reports the top five methods per scenario; full leaderboards are kept in the reproducibility package.

Table 3 and Fig. 3 separate the sparse low-D case from the larger-D cases. For $D = 0.5$ with white noise, SNHT leads with mean F1 0.725, followed by Chow at 0.645; the strongest listed learned detector is far below this range. SNHT also has low matched-event localization error, with MAE 1.360 timesteps. Bootstrap confidence intervals are provided in Supplementary Table S1 and do not alter the qualitative regime-level interpretation. A supplementary F1–MAE diagnostic plot is provided in Supplementary Fig. S1. For $D = 1.0$, learned window-based detectors rank higher: Transformer leads in the white-noise and pink-noise scenarios, although the white-noise Transformer–LSTM gap is qualified by the paired test below. For $D = 1.5$ with white noise, Transformer leads under the primary 25-step margin, while LSTM and CUSUM are close. The matching-margin robustness check shows that this ranking is tolerance-sensitive: CUSUM overtakes Transformer when the evaluation tolerance is widened to 50 or 100 timesteps.

The paired tests in Table 4 support the bounded ranking claims. SNHT ex-

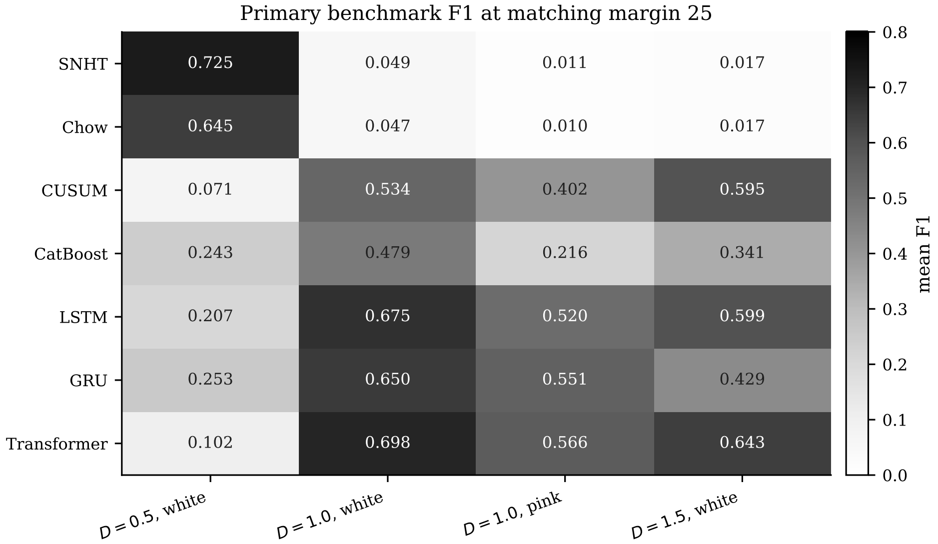

Fig. 3. Primary benchmark F1 at the matching margin of 25 timesteps for the manuscript methods and frozen test scenarios. Classical tests lead in the sparse low-D setting; learned window-based detectors rank higher in the $D = 1.0$ and strict $D = 1.5$ evaluations.

Table 4. Paired Wilcoxon comparisons on per-series F1 for key method pairs.

Scenario	A	B	Alt.	Mean F1 diff	p
D=0.5, white	SNHT	Chow	greater	0.080	0.0228
D=0.5, white	SNHT	GRU	greater	0.472	< 0.0001
D=1, white	Transformer	CUSUM	greater	0.164	< 0.0001
D=1, white	Transformer	GRU	greater	0.048	< 0.0001
D=1, white	Transformer	LSTM	two-sided	0.023	0.0937
D=1, pink	Transformer	CUSUM	greater	0.164	< 0.0001
D=1, pink	Transformer	GRU	two-sided	0.015	0.0011
D=1.5, white	Transformer	CUSUM	greater	0.048	< 0.0001
D=1.5, white	Transformer	LSTM	two-sided	0.044	< 0.0001

ceeds Chow and GRU in the $D = 0.5$, white-noise scenario. At $D = 1.0$ with white noise, Transformer exceeds CUSUM and GRU, but the Transformer–LSTM comparison remains close ($p = 0.0937$). Under the primary margin, Transformer exceeds CUSUM for $D = 1.0$ pink noise and $D = 1.5$ white noise; the robustness pass qualifies the latter result because CUSUM improves under wider tolerances. These paired tests are preselected key comparisons and are used as supporting evidence rather than as a family-wise multiple-testing claim; a Holm-adjusted audit

Table 5. Same-noise $D = 1.0$ diagnostic comparison against the best classical baseline on the same 50 test seeds.

Noise	Transformer F1	Classical F1	Best classical method
white	0.586	0.663	CUSUM
pink	0.263	0.464	CUSUM
brownian	0.121	0.259	CUSUM
blue	0.160	0.523	SNHT/Chow
violet	0.078	0.793	SNHT/Chow

table is included in the reproducibility package.

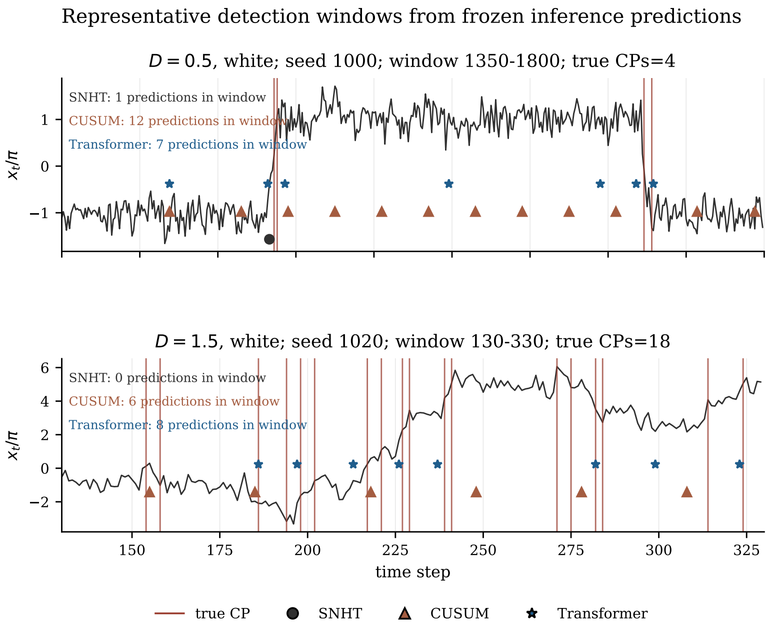

Fig. 4. Representative detection windows: vertical lines mark true change points and markers show selected predictions.

The same-noise experiment in Table 5 gives the colored-noise boundary. This diagnostic is not a replacement for the frozen universal-training benchmark: it uses separately trained noise-family-specific Transformer runs and asks whether this restriction reverses the colored-noise comparison. Training and testing Transformer separately within each $D = 1.0$ noise family still leaves it below the best classical baseline in all five cases. Noise-specific Transformer training alone therefore does not support a universal colored-noise neural-advantage claim under the evaluated pipeline.

7. Discussion

The benchmark should be read as regime-dependent evidence. In the $D = 0.5$ white-noise scenario, labeled events are sparse and comparatively sharp in the assigned state-space level, which aligns with homogeneity and structural-break tests. SNHT and Chow therefore outperform the learned detectors in mean F1, and SNHT also localizes matched events close to the labeled crossing. This is not a general theorem about low-noise benchmarks; it is a result for the evaluated stochastic map, labeling rule, universal-training setup, and matching protocol. At $D = 1.0$ and under the strict $D = 1.5$ margin, local trajectory shape around a crossing becomes more useful, so learned window-based detectors rank higher. The $D = 1.0$ white-noise Transformer–LSTM comparison remains close, and the $D = 1.5$ ranking is tolerance-sensitive. Thus the neural result is best read as a strict-margin event-detection tradeoff, not as a universal claim about transition-time precision.

The colored-noise results limit, rather than broaden, the neural claim. The pink-noise scenario in the frozen benchmark is a transfer check for the universally trained model, while the same-noise diagnostic tests whether noise-specific Transformer training changes the conclusion. It does not: the evaluated Transformer pipeline remains below the best classical baseline across the five tested $D = 1.0$ noise families. For the present FNL framing, this is a useful boundary. Temporal correlation structure is a separate benchmark variable, and detector choice for colored noise should be validated directly rather than extrapolated from the white-noise ranking.

The scope is deliberately narrow. The benchmark is one-dimensional, uses a fixed main trajectory length, and evaluates four frozen scenarios rather than a dense map over D, Δt, and noise families. Its conclusions are strongest for trajectories from the same stochastic map with the same level-crossing labels, split policy, calibration procedure, and matching rule. Alternative state-space partitions, observation noise, latent-state settings, real sampling mechanisms, or different detector calibration rules may change the ranking. These boundaries are also what make the benchmark useful: it gives a controlled reference point for detecting dynamics-induced transition events before moving to less controlled data.

8. Conclusion

This paper presents a reproducible benchmark for comparing change point detectors on dynamics-generated noise-induced transitions. The benchmark fixes the stochastic map, labeling rule, train/calibration/test split, calibration procedure, matching margin, and metrics, so that classical tests and learned detectors are compared under one protocol.

The frozen four-scenario results show D- and noise-dependent rankings. In the sparse $D = 0.5$ white-noise scenario, classical homogeneity tests lead. Under the primary 25-step margin, learned window-based detectors lead the two evaluated

$D = 1.0$ scenarios and the strict-margin $D = 1.5$ white-noise scenario. These conclusions are bounded: the $D = 1.5$ ranking is tolerance-sensitive, and the $D = 1.0$ white-noise Transformer–LSTM gap is not statistically significant in the paired test.

The same-noise diagnostic shows that noise-specific Transformer training alone does not produce a universal colored-noise neural advantage under the evaluated pipeline. The main contribution is therefore the controlled benchmark and the evidence it provides for the evaluated stochastic-map scenarios. Future work should test D- and noise-conditioned detectors, alternative label partitions, and settings with noisy or latent observations.

Declaration of competing interest

The authors declare no competing interests.

Data and code availability

Code, configuration files, and generated benchmark outputs are maintained in the project repository at `https://github.com/Nikkfh5/cpd`. The repository will be archived and assigned a persistent identifier upon acceptance.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

[1] R. P. Adams and D. J. C. MacKay, “Bayesian online changepoint detection”, 2007, arXiv:0710.3742.
[2] S. Aminikhanghahi and D. J. Cook, “A survey of methods for time series change point detection”, *Knowledge and Information Systems* **51** (2017) 339–367.
[3] C. Truong, L. Oudre and N. Vayatis, “Selective review of offline change point detection methods”, *Signal Processing* **167** (2020) 107299.
[4] G. J. J. van den Burg and C. K. I. Williams, “An evaluation of change point detection algorithms”, 2020, arXiv:2003.06222.
[5] R. Killick, P. Fearnhead and I. A. Eckley, “Optimal detection of changepoints with a linear computational cost”, *Journal of the American Statistical Association* **107** (2012) 1590–1598.
[6] P. Fryzlewicz, “Wild binary segmentation for multiple change-point detection”, *The Annals of Statistics* **42** (2014) 2243–2281.
[7] S. Gharghabi, C.-C. M. Yeh, Y. Ding, W. Ding, P. Hibbing, S. LaMunion, A. Kaplan, S. E. Crouter and E. Keogh, “Domain agnostic online semantic segmentation for multi-dimensional time series”, *Data Mining and Knowledge Discovery* **33** (2019) 96–130.
[8] A. Ermshaus, P. Schäfer and U. Leser, “ClaSP: Parameter-free time series segmentation”, *Data Mining and Knowledge Discovery* **37** (2023) 1262–1300.
[9] A. Ryzhikov, M. Hushchyn and D. Derkach, “Latent stochastic differential equations for change point detection”, *IEEE Access* **11** (2023) 104700–104711.

[10] H. A. Kramers, "Brownian motion in a field of force and the diffusion model of chemical reactions", *Physica* **7** (1940) 284–304.

[11] M. I. Freidlin and A. D. Wentzell, *Random Perturbations of Dynamical Systems* (Springer, New York, 1998), 2 edition.

[12] N. Berglund and B. Gentz, *Noise-Induced Phenomena in Slow-Fast Dynamical Systems: A Sample-Paths Approach* (Springer, London, 2006).

[13] V. S. Anishchenko, I. A. Khovanov, N. A. Khovanova, D. G. Luchinsky and P. V. E. McClintock, "Noise-induced escape from the lorenz attractor", *Fluctuation and Noise Letters* **1** (2001) L27–L33.

[14] D. Valenti, A. Fiasconaro and B. Spagnolo, "Role of the colored noise in spatio-temporal behavior of two competing species", *Fluctuation and Noise Letters* **5** (2005) L337–L342.

[15] L. O. Barauna, R. R. Rosa, C. A. Wuensche, R. A. Sautter, V. A. de Santiago Júnior, E. H. Shiguemori and M. B. Padua, "Characterizing colored noise time series patterns with deep learning models", *Fluctuation and Noise Letters* **23** (2024) 2450053.

[16] E. S. Page, "Continuous inspection schemes", *Biometrika* **41** (1954) 100–115.

[17] A. N. Pettitt, "A non-parametric approach to the change-point problem", *Applied Statistics* **28** (1979) 126–135.

[18] H. Alexandersson, "A homogeneity test applied to precipitation data", *Journal of Climatology* **6** (1986) 661–675.

[19] T. A. Buishand, "Some methods for testing the homogeneity of rainfall records", *Journal of Hydrology* **58** (1982) 11–27.

[20] G. C. Chow, "Tests of equality between sets of coefficients in two linear regressions", *Econometrica* **28** (1960) 591–605.

[21] Z. Harchaoui, E. Moulines and F. R. Bach, "Kernel change-point analysis", in *Advances in Neural Information Processing Systems* (2009), volume 21, pp. 609–616.

[22] S. Liu, M. Yamada, N. Collier and M. Sugiyama, "Change-point detection in time-series data by relative density-ratio estimation", *Neural Networks* **43** (2013) 72–83.

[23] J. Li, P. Fearnhead, P. Fryzlewicz and T. Wang, "Automatic change-point detection in time series via deep learning", *Journal of the Royal Statistical Society Series B: Statistical Methodology* **86** (2024) 273–285.

[24] J. Timmer and M. Koenig, "On generating power law noise", *Astronomy and Astrophysics* **300** (1995) 707–710.

[25] N. J. Kasdin, "Discrete simulation of colored noise and stochastic processes and $1/f^{\alpha}$ power law noise generation", *Proceedings of the IEEE* **83** (1995) 802–827.